Chapter 4: Nowcasting Mixed-Frequency Bitemporal High Priority Macroeconomic Time Series

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1 Introduction

This document details on asset specific data infrastructure specifically to handle only data engineering to serve directional mid frequency U.S. Equity forecasting. Economic variables exist in a complex system that is hard to determine using a closed function, even running Conditional Independence Tests. Due to the instability of Conditional Independence Tests to produce the similar graphs due to the conditioning set property, Variable Auto Regression for forecasting using weight lag edges derived from Conditional Independence Tests was dropped and substituted for a Nowcasting Kalman Filter.

2 Dataset from Federal Reserve Bank of St. Louis

Macroeconomics studies economy wide variables such as Gross Domestic Product (GDP), national unemployment rates, inflation (CPI), central bank interest rates, and national debt. Microeconomics focuses on individual economic units, such as a supply and demand in an isolated market. The Federal Reserve Bank of St. Louis created FRED (Federal Reserve Economic Data) primarily to aggregate high level economic indicators collected by government agencies (e.g., the Bureau of Labor Statistics, Bureau of Economic Analysis) and central banks. Because its primary data series describe the economy as a whole, it is a macroeconomic database. The FRED database has a total of 820k series. The series can be grouped and parsed using their category id tag, which is a hierarchy file system to group series by their respective fields.

series_id character varying	observation_start date	observation_end date	frequency_short character varying	units_short character varying	seasonal_adjustment character varying	popularity integer	title character varying
DGS10	1962-01-02	2026-07-24	D	%	Not Seasonally Adjusted	99	Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity, Quoted on an Investment Ba:
T10Y2Y	1976-06-01	2026-07-27	D	%	Not Seasonally Adjusted	99	10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity
DFII10	2003-01-02	2026-07-24	D	%	Not Seasonally Adjusted	92	Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity, Quoted on an Investment Ba:
T10Y3M	1982-01-04	2026-07-27	D	%	Not Seasonally Adjusted	92	10-Year Treasury Constant Maturity Minus 3-Month Treasury Constant Maturity
T10YIE	2003-01-02	2026-07-27	D	%	Not Seasonally Adjusted	90	10-Year Breakeven Inflation Rate
GFDEBTN	1966-01-01	2026-01-01	Q	Mil. of $	Not Seasonally Adjusted	86	Federal Debt: Total Public Debt
DFF	1954-07-01	2026-07-24	D	%	Not Seasonally Adjusted	86	Federal Funds Effective Rate
VIXCLS	1990-01-02	2026-07-27	D	Index	Not Seasonally Adjusted	86	CBOE Volatility Index: VIX
DGS5	1962-01-02	2026-07-24	D	%	Not Seasonally Adjusted	86	Market Yield on U.S. Treasury Securities at 5-Year Constant Maturity, Quoted on an Investment Basi

Figure 1: Series Dataset

A **FRED Macroeconomic Series's** series id (see figure 1) is the hash that one uses to access the series inside the database. The observation start, end is the date of the earliest and latest data point in the FRED database. The frequency is the expected time between 2 data points in a series. The units is the value that should scale the base number in the FRED database. The seasonal adjustment determines if a series is or isnt treated by a seasonality algorithm. The title is the description of the series for humans to read. The popularity score is the most up to date current traffic score using a hidden algorithm by FRED information technologies team.

	series_id character varying	date date	realtime_start date	realtime_end date	value double precision
1	DGS10	2026-05-29	2026-06-01	9999-12-31	4.45
2	DGS10	2026-05-28	2026-05-29	9999-12-31	4.45
3	DGS10	2026-05-27	2026-05-28	9999-12-31	4.48
4	DGS10	2026-05-26	2026-05-27	9999-12-31	4.5
5	DGS10	2026-05-25	2026-05-27	9999-12-31	[null]
6	DGS10	2026-05-22	2026-05-26	9999-12-31	4.56
7	DGS10	2026-05-21	2026-05-22	9999-12-31	4.57
8	DGS10	2026-05-20	2026-05-21	9999-12-31	4.57

Figure 2: Data Series for DGS10, Market yield on US Treasuries at 10 Year Constant Maturity

A **FRED Macroeconomic Observation** can be described using a date, realtime start date, and value. The date, value is straight forward, the value that is associated with the date. The realtime start is the earliest time that a data point was available. The series DGS10 has a 2 day delay between the date, and the realtime start, as seen in figure 2. This difference in observation time and realtime start time is known as latency or lag. A observation could be revised, which means that for a unique pair of series id, date could return a unique realtime start and value. This many (realtime start, value) to one (series id, date) relationship describes the bitemporal property of macroeconomic series. In short, this describes "what was known back then vs. what is known back then from today's point of view".

2.1 Fakeness of The Popularity Score

In addition to structural categorization, each series carries a discrete popularity score $P \in \{0, \dots, 100\}$. This score reflects real-time user traffic and API endpoint request density. As illustrated in Figure 3, the popularity metric follows a right-skewed distribution resembling a binomial process, with a empirical mean of $\mu = 1.69$ and a mode at $\text{Mode} = 1.0$. This distribution although a subset (150 thousand samples) of the series relevant to U.S. Equity Mid Frequency pricing models.

Because the popularity score does not exist as a timeseries, a policy that derives a set via an arbitrary cutoff score is both arbitrarily in threshold and irreproducible historically. Secondly, popularity of a series does not entail its predictive efficacy, and conversely should decrease the predictive edge for any profitable strategies via the Efficient Market Hypothesis (EMH) Fama [1970], as any edge that it produces would be traded away. Thus, any selection policy derived from popularity must suffer from both lack of historical reproducibility and thus introduces survivorship bias and lack of profitable edge in deployment.

2.2 Minor Advantage of Archaic Data Hierarchies

Given the unreliability of filtering series purely by popularity metrics or pre-adjusted flags, classification must rely on structural category hierarchies. These hierarchies can be modeled via graph search algorithms—such as Breadth-First Search (BFS) or Depth-First Search (DFS)—starting from the root directory.

The legacy file architectures of databases like FRED exhibit a many-to-one mapping, where an individual series may belong to multiple subcategories. This legacy structure offers a minor empirical advantage, that categories remain stable over time due to institutional inertia. Sta-

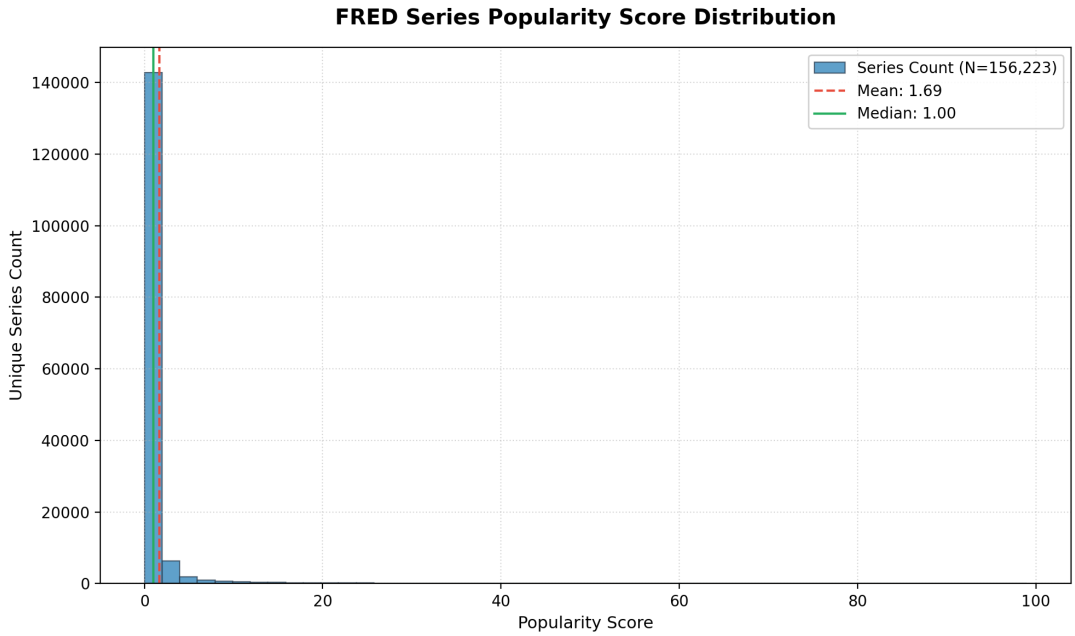

Figure 3: Empirical distribution of Subset of Relevant FRED series popularity scores showing extreme right-skewness ($\mu = 1.69, \text{Mode} = 1.0$).

tistical agencies rarely possess the bandwidth or administrative authorization to re-categorize legacy series.

3 Data Pruning

3.1 Data Viability Under Fixed Feature Set Size Constraint

As of 2026/08/06, we were unable to design a stable form of fixed size compression for a dynamic sized time series dataset. Thus we dropped that and returned to fixed sized feature set.

In a Mid Frequency U.S. Equity Modeling, not all data is useful. Consider a Dataset with series A, B, C, D, E where A, B start, end is both 1990 to current day, while C starts at 1990 to 2000, D starts at 1995 to current day, and E starts at 2025 to current day. C is useless for a pricing model because we cannot derive new observations, thus it is thrown out. Series E is useless because we cannot systematically and deterministically back test it far enough to fundamentally compute the usefulness of the indicator. D which starts at 1995, can be backtestable, but to preserve the fixed size feature set A, B would have to be truncated to the start of D or throw out D. This decision is determined by the importance of D weighed against the importance of the truncated history of A, B and the other sets.

If a series does not offer data before the start of backtest and also is currently active, it is excluded from the viable dataset. While this introduces selection bias and survivorship bias, this bias is unavoidable because the alternative is to have a random sampled via point in time selection by availability and thus yielding a varied feature set which makes training impossible throughout time and produces weak signals due to the possibility of sampling a low signal series.

An Economic theory on why D case is rare given the backtesting date is greater than the intialization of the underlying database:

1. The final set is terribly derived and an indicator is not actually a leading or influential indicator, and thus can be safely removed/replaced with more leading indicator, and thus it cannot be the D case.

2. A genuinely important data points just released after the intended start of backtest, abnormally rare, and thus must make tradeoff here.

So because it is so rare that such a case exists, there is little risk of truncating backtest to too small of time samples, granted that the backtesting starts from 2015.

3.2 Semantic Filtering

To maintain a fixed, parsimonious parameter space across all U.S. equities in a unified mid-frequency pricing model, feature selection must strictly prioritize market-wide, systemic signals while eliminating localized noise and redundant metrics.

When predicting equity price distributions, candidate economic time series are filtered based on three intuitive criteria:

1. **Elimination of Micro-Local and Demographic Noise:** Indicators tied to specific geographic sub-regions (e.g., state-level metrics) or narrow demographic slices (such as age, race, educational attainment, or veteran status) are pruned. While relevant to localized economic studies, these micro-variables add localized variance without providing systemic informational value for broad equity pricing.

2. **Removal of Non-Stationary and Derived Transformations:** Derived metrics representing growth rates, percentage changes, or chained units are excluded in favor of fundamental, raw causal drivers. Unbounded percentage shifts introduce structural non-stationarity and extreme collinearity into the feature matrix, degrading model stability during training.

3. **Pruning Institutional and Structural Redundancy:** Data series representing disaggregated accounting breakdowns—such as specific bank balance-sheet subsets, localized tax collections, or niche sectoral filings—are omitted. Because broader macroeconomic aggregates (e.g., policy rates, broad money supply, market-wide liquidity indicators) encompass the explanatory power of these lower-tier series, excluding subordinate indicators drastically compresses the feature space without compromising predictive scope.

3.3 The Fakeness of Seasonally Adjusted Series

A critical issue in macroeconomic modeling is the survivorship and selection bias inherent in curated datasets, such as those maintained by FRED economists. There is no guarantee that all time series are seasonally adjusted or smoothed uniformly, leading to a natural oversampling of popular series. However, popularity does not imply predictive utility or leading-indicator status.

When FRED economists perform seasonal adjustment, they typically rely on the **X-13ARIMA-SEATS** algorithm Monsell [2007].

3.3.1 The X-13ARIMA-SEATS Algorithm

The X-13ARIMA-SEATS procedure decomposes an observed non-stationary time series Y_t into four unobserved components: trend-cycle (T_t), seasonal (S_t), trading day/holiday effects (I_t), and irregular variations (ϵ_t).

Depending on the nature of the series, the decomposition is expressed in either **multiplicative** or **additive** form:

$$\text{Multiplicative Decomposition: } Y_t = T_t \times S_t \times I_t \times \epsilon_t \tag{1}$$

$$\text{Additive Decomposition: } Y_t = T_t + S_t + I_t + \epsilon_t \tag{2}$$

The underlying mechanism relies on a RegARIMA (Regression with ARIMA errors) model to model and pre-adjust the series prior to seasonal filtering:

$$\phi_p(B)\Phi_P(B^s)(1-B)^d(1-B^s)^D(Y_t - \sum_k \beta_k X_{kt}) = \theta_q(B)\Theta_Q(B^s)a_t \tag{3}$$

where:

- B is the backshift operator ($BY_t = Y_{t-1}$).
- s represents the seasonal period (e.g., $s = 12$ for monthly data).
- $\phi_p(B)$ and $\theta_q(B)$ are non-seasonal AR and MA polynomials of orders p and q.
- $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are seasonal AR and MA polynomials of orders P and Q.
- $(1-B)^d$ and $(1-B^s)^D$ denote non-seasonal (d) and seasonal (D) differencing operators.
- X_{kt} represents exogenous regression variables (e.g., trading day, Easter, outliers).
- $a_t \sim \mathcal{N}(0, \sigma^2)$ is a white noise process.

Once the RegARIMA model extends the series at both ends via backcasts and forecasts, signal extraction filters (either X-11 moving averages or SEATS model-based Wiener-Kolmogorov filters) are applied across the entire extended history to extract S_t.

3.3.2 Limitations in Predictive Contexts

Problem 1: Lookahead Bias and Revisions. Despite its effectiveness in smoothing historical series for descriptive analysis, the ARIMA component operates across the full temporal span of the dataset. This creates severe lookahead bias when constructing out-of-sample models. True real-time forecasting requires point-in-time point reconstructions at every backtest node. Even when working with point-in-time vintages, the non-local nature of two-sided filtering creates instability and a lack of reproducibility for extrapolation.

Problem 2: Encrypted Ciphers. While the X-13ARIMA-SEATS software is widely distributed across Python, Excel, and R, the actual parameter settings used by statistical agencies to produce official smoothed series are often unpublished. Because these implicit parameter configurations (”ciphers”) are unavailable, converting official seasonally adjusted values back to their raw counterparts (or vice versa) in a deterministic manner becomes practically impossible.

Conclusion. While smoothed data is useful, due to the lack of availability of ciphers of the underlying algorithm parameter values, not seasonally adjusted series are more preferable to perform modeling, as we can guarantee that the point in time seasonally adjusted values are feasibly reconstructable with respect to their point in time not seasonally adjusted values.

3.4 Proposed Occam's Razor Causal Pruning

A classic example of a spurious correlation is the relationship between ice cream sales and crime rates, which is driven by an underlying confounder: temperature.

> As ice cream sales increase, violent crime rates also rise. However, this correlation is non-causal and driven by a hidden third variable: warm weather. Higher summer temperatures cause both an increase in ice cream consumption and an increase in outdoor social interactions, thereby elevating the statistical likelihood of violent crime.

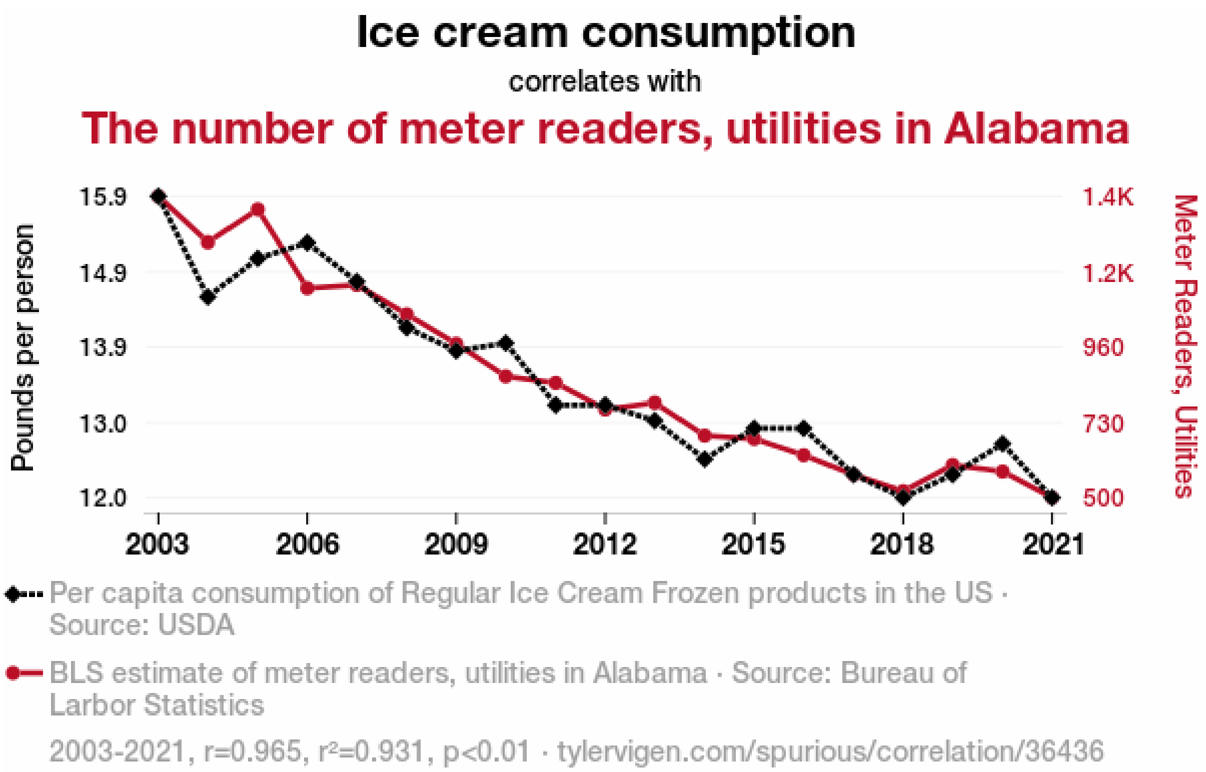

Figure 4: Example of Spurious Correlation between Ice Cream Sales and Crime Rates.

If temperature independently drives both ice cream sales and crime rates, we can model this three-variable system by designating the confounder (Temperature) as the parent node to both child variables (Crime, Ice Cream Sales).

In time series data, this causal relationship manifests as a temporal lag: a shift in temperature at time t leads to a subsequent shift in ice cream sales and crime rates at $t + \tau_{\text{ice cream}}$ and $t + \tau_{\text{crime}}$, respectively. Because temperature is both **causal and leading**, it can be used to forecast future values of both child variables.

3.4.1 Causal Propagation and Feature Pruning

Now consider a downstream child variable caused by ice cream sales, such as diabetes rates. Because ice cream consumption is causal and leading relative to diabetes rates, it can be used as a predictive feature.

By transitivity, temperature must also be able to predict diabetes rates—albeit with a longer temporal lag (τ) and higher forecast error—because it is a leading cause of ice cream consumption. This yields two key structural properties:

1. **Expressiveness & Scope:** Temperature is broader in scope and higher in the causal hierarchy, exerting influence over a wider network of downstream variables than ice cream sales alone.

2. **Redundancy:** If temperature is present in the dataset, ice cream sales becomes redundant for predicting diabetes rates and can be safely omitted.

Occam's Razor Causal Pruning: If a variable X is both more leading (larger temporal precedence) and broader in scope (higher causal influence) than variable Y relative to a target series Z, X is strictly more expressive than Y. Consequently, Y can be pruned without losing fundamental causal information.

Application to Financial Models: For a fixed-parameter model for mid-frequency U.S. Equity pricing, we must prune niche or subordinate indicators. Applying Occam's Razor Causal Pruning prevents parameter explosion and eliminates extreme multicollinearity within the feature space.

3.5 Extended Occam's Razor Causal Pruning

If the Occam's Razor Causal Pruning is acceptable, then the same property must be extended to almost every alternative data source. To predict a retail warehouse company's revenue like Walmart, whilst Modern trading companies use traffic data or satellite imagery of parking lots, it is the cause which is Real Disposable Personal Income (RDPI) that dictate whether or not a person can afford to shop at Walmart using their portion of disposable income. And although RDPI has publication latency, if we construct the causal parent of RDPI, or the parent of RDPI, such as U.S. Exports, we can effectively simulate having satellite imagery by connecting higher economic output yields higher GDP, which increases government income, which increases rate that government gives stipend/benefits which increases Disposable income. Thus satellite imagery is practically useless for macroeconomic mid frequency pricing models, because it does not generalize across all assets. Similarly, we would also not need traffic, because that is dependent on gas prices, which is dependent on the oil production and exports, in which we would just use futures price of oil to simulate traffic data. Thus this would also render most of real estate and land prices lagging, and thus useless. This property would also render most of agriculture, most news articles, government policy lagging and thus useless. Futhermore this would also render most of the dataset on fred lagging, and thus useless. Thus the set of leading variables are the forward looking variables such as commodities futures, tradeable options, and the set of leading macroeconomic indicators. And since company revenue are lagging and caused by leading indicators, and by the empirical property in Chapter 2 on the soft bounded distribution of stock prices to a multiple of a company's fundamental value which is derived from its revenue, it follows that there must exist some set of leading indicators that predict stock price.

Thus for mid frequency U.S. Equity under one pricing model with the restriction of common and fixed parameter space, to reduce parameter size and colinearity for stable training, we derive the following properties which construct the Extended Pruning Theory:

1. Lagging indicators are less preferable to leading indicators.
2. Annual indicators are slow because they trail over the entire year, they are unchanging thus the tradeoff of parameter space to signal strength is not worth it to preserve annual frequency indicators. In other words because annual indicators publish too infrequently, it is both laggy and less statistically significant.
3. Indicators that were and still are revised in high magnitude in either time elapsed or value are considered lagging. Thus it is more preferable to have the indicator that gets less revised given 2 indicators with the same leadingness and influence.
4. Indicators that are too low in the chain are do not generalize equivalent level of relevance across all assets, and thus are useless in a fixed parameter space mid-frequency pricing model.

Unfortunately not all criteria can be met, and again, tradeoffs need to be made to accommodate deriving a set of true parents.

3.6 Nowcasting Over Forecasting

If the Extended Pruning Theory is acceptable, then the set of ideal indicators would be both leading and influential. For the subset of the resulting set, if we had selected the most leading and the most influential, we inevitably end up with a set of true parents, variables that do not have parents, thus they cannot be predicted. An example is executive order to start war or the FFR. Because of this, there is no reason to forecast, because we cannot forecast values that do not have causal parents and have not materialized. Thus we can reduce a forecasting problem into a nowcasting problem, where we take the observed values, and only solve for the latent intermediate values in between the publication lag. Thus due to the existence of high frequency series, for example commodity futures, FFR, or volatility futures, their latency is often around 1 day, while the monthly series latency is 1-2 months (see figure 5), we can use the higher frequency leading indicators to bring up to date (nowcast) the lower frequency leading indicators using a fitted Kalman filter.

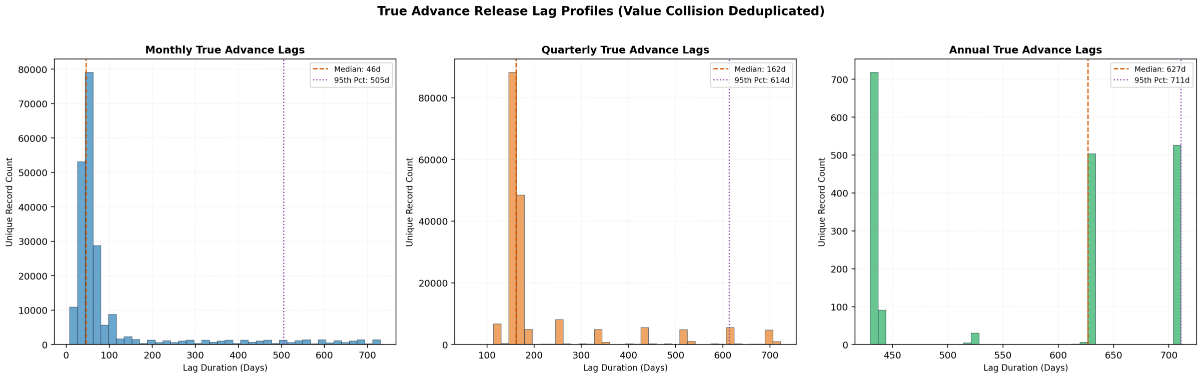

Figure 5: First Publication Latency

3.7 Necessity of Non-Quantitative Indicators and its Risks

While tradeable quantitive indicators such as commodity, volatility, indice futures are leading, they suffer from high noise and require high liqudity to be reliable. Non-Tradeable quantitative economic indicators like FFR, Payroll, and Producer Price Index are causal parents, but they are slow in publication, around 1-2 month publication latency. Furthermore, these indicators are not immediately reflect rapid regime changes for example, the president declared war on Iran, which influenced the global supply of oil, petrol, and plastic products. Thus some news articles (specifically regarding war and potential demand/supply chain shock) are necessary to derive rapid structural changes, despite that commodity future most likely have already reflected them (however noisily) and serve as a noisy proxy.

To process the news using a LLM comes with a drawback. Given any data source, such as a news article, we want to confine it to a fixed feature space, thus we want the classification token from the LLM, which is a fixed size vector determining the sentiment of the inputted text. However the LLM that is used to read the article has training data that is not available point in time of the simulation, thus using a LLM will have minor look forward bias, thus the backtest will almost never experience a semantic black swan event, and thus it is risky to assume backtesting results will carry over robustness to black swan events in deployment given that a LLM trained on modern data was used.

3.8 Conditional Independence Fails to Produce Stable Edges, thus Dynamic Sized Graph Embeddings Failed

Historically, Granger causality testing Granger [1969] has served as the baseline statistical framework for identifying lead-lag dynamics in temporal data. However, bivariate Granger frameworks are fundamentally limited when deployed in complex, highly dynamic economic systems. Because Granger causality relies on pairwise autoregressive comparisons, it fails to account for multivariate confounding factors or simultaneous feedback loops. While cyclic feedback systems can be mathematically isolated by abstracting the global graph topology into a Directed Acyclic Graph (DAG) of Strongly Connected Components (SCCs) to model the macro-level flow of systemic influence (see figure 6), standard linear frameworks remain inadequate for high-dimensional, time-varying contexts.

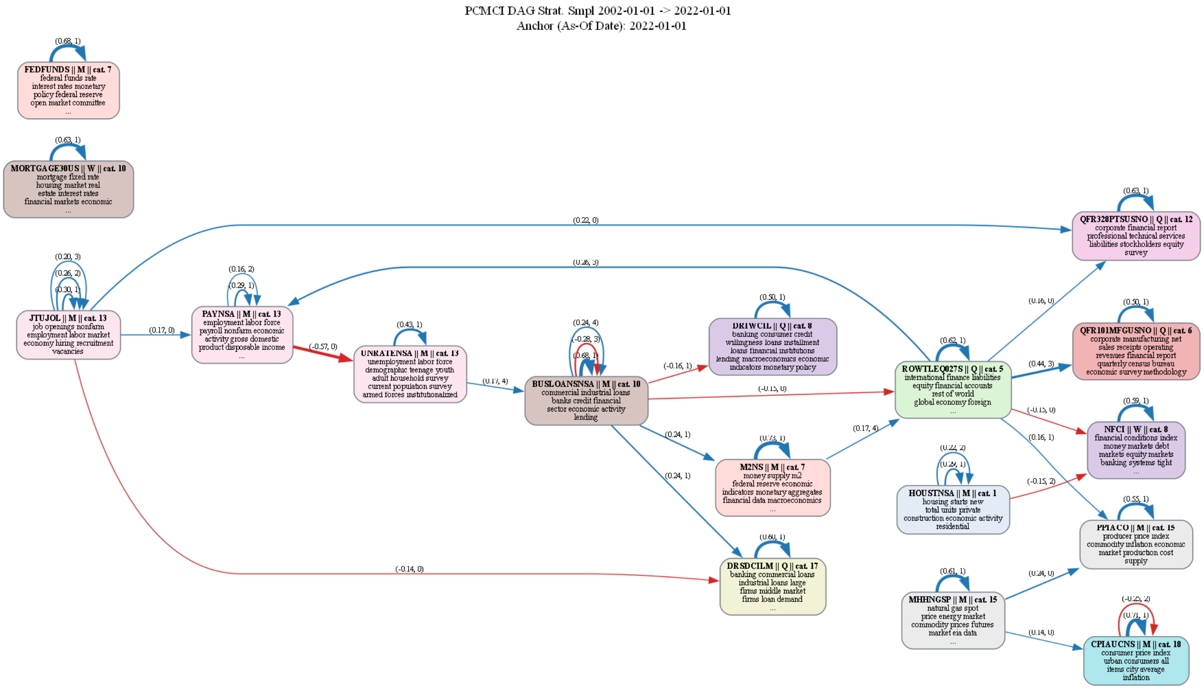

Figure 6: Causality Graph Via Joint Peter Clark Momentary Conditional Independence (PCMCI) Algorithm

To resolve these structural complexities, contemporary paradigms leverage constraint-based causal discovery. The foundational PC algorithm Spirtes [2000] Zheng et al. [2023] uses conditional independence tests to map observational distributions into causal DAGs. More recently, the PCMCI framework Runge [2022] advanced this paradigm by introducing conditional momentary independence (CMI) tests. By explicitly conditioning on past slices of the time series up to a maximum lag τ, PCMCI robustly isolates true time-lagged causal links from autocorrelation artifacts, providing both the directed edge topology and its respective structural transmission weight.

While deploying constraint-based discovery over a localized set of indicators is mathematically trivial, scaling this architecture to encompass the global macroeconomic universe introduces two critical engineering bottlenecks:

1. **Computational Intractability and Scaling Limits:** The combinatorial search space for conditional independence testing scales exponentially with the number of variables D. Empirically, processing asset arrays exceeding $D > 500$ nodes becomes computationally unfeasible, with single walk-forward rolling estimations requiring hours of execution time. For live inference and real-time execution, this latency is unacceptable, necessitating architectures that enforce structural persistence or structural compression.

2. **Dimensional Instability in Unconstrained Spaces:** Real-world macro-data matrices exhibit irregular input dimensional spaces. While a core set of persistent equilibria indicators exists (e.g., Gross Domestic Product, Consumer Price Index), introducing peripheral series frequently results in severe input sparsity or architectural failures when edges are

hard-filtered by statistical strength. Consequently, standard models lack the capacity to accept an unconstrained, variable-length input space without structural collapse.

Recent advancements in amortized causal inference, such as AVICI Lorch et al. [2022] and CausalPFN Balazadeh et al. [2025], attempt to resolve variable dimensional constraints by utilizing permutation-invariant and permutation-equivariant attention layers over the feature dimension. Theoretically, this allows a frozen neural network to process inputs of varying dimensional sizes (D) without structural reconfiguration.

However, these amortized foundation models suffer from a severe architectural ceiling in real-world deployments. First, they execute attention mechanics directly across the variable axis, incurring a quadratic computational complexity of $\mathcal{O}(D^2)$ that triggers catastrophic out-of-memory (OOM) failures on standard hardware accelerators when scaled toward the scale of a true economy ($D \approx 10,000$). Second, these frameworks are trained exclusively on synthetic data simulators generating idealized mathematical graphs. As a result, their out-of-distribution generalization collapses when confronted with the highly specialized, non-linear structural motifs of empirical financial networks.

Real-world macroeconomic data arrays differ fundamentally from the unannotated tabular domains targeted by existing literature. FRED engine provide dense textual metadata, detailed semantic definitions, and structured indexing tags alongside raw numeric series. While a theoretical architecture is capable of translating global textual features into computable causal embeddings, it violates the principle of reducing a feature space to the minimum set of true causal parents that cannot be forecasted in itself. Divide and conquer approach by decomposing a large causal space into subsets of timeseries with a joint set of shared exgoneous variables via causal independence algorithms still produce unstable relationships across batches despite sharing 75% of the same variables in both small and large variable set. In small variable sets, the pcmci stability across datasets with a shared joint set suffer from first the omission of confounders, and secondly vulnerability of conditioning set size truncation, which truncates the search space via early detection of conditional independence, which compounds into a unstable graph even with the introduction of a relatively small explore set. In large sets, conditional independence checks get overwhelmed with the raw datasize and thus is unavoidable to deduce false positives in conditional independence, similar to spurious correlation, this end up producing nonsensical causal links that again compounds into unstable graphs despite sharing a joint exogeneous set and having a small explore set. Finally we cannot guarantee that the signal is preserved in a black-box attention network. Due to these constraints, conditional independence was not used to autoregress forecasts, and we swapped out for a nowcasting algorithm. Finally, smoothing and stationarity of time series was naive and subject to unstability at the time of development of causal inference, thus this component failed to materialize due to the challenges faced in Mixed Frequency and Non stationary time series at the time. However if revisited with knowledge as of 08/06/2026, it is possible that this component is viable with the newer techniques in handling mixed frequency, time series stationarity, and swapping target from forecasting to nowcasting.

4 Bitemporal Properties

Macroeconomic indicators do not update continuously. Unlike financial asset prices, which stream in real time, macroeconomic series suffer from significant publication lags and ongoing agency revisions. Consequently, simple forward-filling introduces severe lookahead bias into historical feature matrices.

To capture the true information state available to a market participant at any given point in time t, each observation must be indexed by two distinct temporal dimensions:

1. **Observation Date (t_{obs}):** The underlying period being measured (e.g., Q1 2024 or January 2025).

2. **Realtime Date ($t_{realtime}$):** The exact calendar timestamp when the specific data point or revision was published by the reporting agency.

Evaluating a point-in-time slice as of timestamp T_{now} requires filtering the historical database under a strict dual inequality:

$$\mathcal{D}(T_{now}) = \{(t_{obs}, x) \mid t_{obs} \leq T_{now} \quad \text{AND} \quad t_{realtime} \leq T_{now}\} \tag{4}$$

By sequentially evaluating this dual constraint across historical timesteps, we quantify the total volume of agency revisions. As shown in Figure 7, revision intensity varies drastically across series: while certain stable series exhibit zero post-release changes, core macroeconomic drivers undergo frequent historical revisions over multi-year windows.

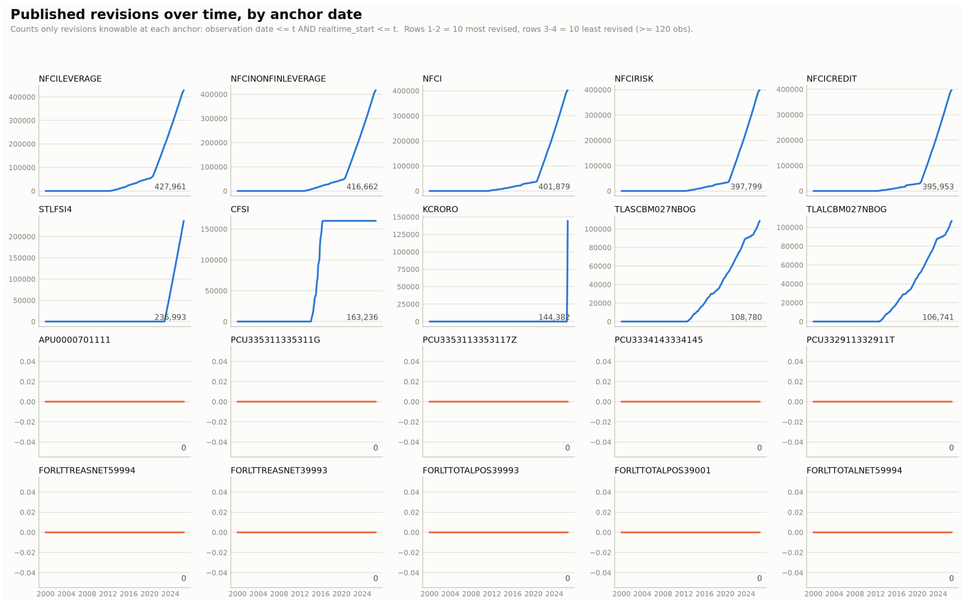

Figure 7: Cumulative revision history across the filtered series subset. Top panel: The 10 most frequently revised macroeconomic series (e.g., Nonfarm Payrolls, GDP). Bottom panel: The 10 least revised series, exhibiting near-zero post-release adjustments.

4.1 The States of a Bitemporal Variable

At any given evaluation stride τ, every target tuple (i, t) exists in one of three distinct structural lifecycle states:

1. **Pre-Release ($t > \tau - \delta_i$):** The reference period t is completely unobserved due to publication latency δ_i. The model nowcasts the value using the state-space factor trajectory:

$$\hat{y}_{i,t|\tau} = \lambda_i^\top \hat{\mathbf{f}}_{t|\tau}$$

2. **Post-Release Vintage ($\tau - \delta_i \leq \tau_{\textbf{release}} \leq \tau$):** The reporting agency publishes the initial official figure $y_{i,t,\tau}^{(1)}$. For series subject to ongoing revisions, this initial print acts as a noisy proxy containing measurement error:

$$y_{i,t,\tau}^{(1)} = y_{i,t,\infty} + e_{i,t,\tau}, \quad e_{i,t,\tau} \sim \mathcal{N}(0, \sigma_{\text{rev},i}^2)$$

   Series like `CPIAUCNS` have zero revision variance ($\sigma_{\text{rev},i}^2 = 0.00$), rendering their initial release terminal. Conversely, indicators like financial conditions (`NFCI`, $\sigma_{\text{rev},i}^2 = 2.11$) or housing starts (`HOUSTNSA`, $\sigma_{\text{rev},i}^2 = 0.13$) exhibit significant initial release noise.

3. **Oracle ($y_{i,t,\infty}$):** The fully revised, converged ground-truth figure observed after all benchmark adjustments and annual seasonal re-estimations have settled ($\tau \to \infty$).

Table 1: Revision Variance and Publication Latency Metrics (1MS Resample Frequency)

Series ID	Rev Var (Scaled)	Latency Mean (Months)	Latency Std (Months)
CCNSA	0.017853	0.42	0.16
CFNAI	0.177496	1.75	0.15
CPIAUCNS	0.000000	1.36	0.07
DFF	0.000000	0.06	0.03
DGS10	0.000000	0.05	0.03
DGS2	0.000000	0.05	0.03
DHHNGSP	0.000000	0.22	1.85
DRTSCILM	0.000000	1.15	0.11
DTWEXBGS	0.001947	0.17	0.05
GASREGCOVW	0.000000	0.01	0.02
GFDEBTN	0.000004	5.03	0.20
HOUSTNSA	0.134071	1.55	0.05
ICNSA	0.043356	0.19	0.16
JTUJOL	0.103519	2.11	0.13
NFCI	2.111575	0.17	0.03
PAYNSA	0.020333	1.15	0.20
PPIACO	0.092461	1.41	0.19
RAILFRTINTERMODAL	0.012769	2.61	0.43
RSAFSNA	0.013149	1.50	0.17
STLFSI4	0.199130	0.50	0.00
T10Y3M	0.000000	0.00	0.00
TLBACBW027NBOG	0.147190	0.30	0.01
TOTBKCRNSA	0.133229	0.30	0.01
UMCSENT	0.000261	0.85	0.10
UNRATENSA	0.000000	1.15	0.20

4.2 Empirical Analysis of Latency and Revision Variance

The parameters summarized in Table 1 reveal significant cross-sectional heterogeneity in how information enters the state-space model, explaining why naive persistence fails for specific asset classes:

4.2.1 Publication Latency Extremes (δ_i)

Daily market rates such as Federal Funds (`DFF`) and Treasury yields (`DGS10`, `DGS2`) exhibit near-zero latency ($\delta_i \approx 0.05$ months), making their initial values available almost instantaneously. Conversely, structural series face severe reporting delays. Public debt (`GFDEBTN`) suffers from an average lag of 5.03 months, while rail freight traffic (`RAILFRTINTERMODAL`) and job openings (`JTUJOL`) lag by 2.61 and 2.11 months, respectively. For these long-lagged series, a forward-fill strategy forces the model to look multiple quarters into the past. The dynamic factor model achieves significant forecast skill on these series by using zero-lag financial series to bridge the multi-month gap.

4.2.2 Revision Variance Dynamics ($\sigma^2_{\textbf{rev}}$)

Series fall into two structural regimes regarding revision volatility:

- **Terminal Initial Releases ($\sigma^2_{\textbf{rev}} = 0.00$):** Series such as consumer prices (`CPIAUCNS`), unemployment (`UNRATENSA`), and Treasury yields do not undergo historical revisions after release. For these series, the model's predictive edge stems entirely from estimating unobserved current-period values prior to publication date.

- **High-Noise Initial Releases ($\sigma^2_{\textbf{rev}} > 0.10$):** Financial condition indices (`NFCI`, $\sigma^2_{\text{rev}} = 2.111$), housing starts (`HOUSTNSA`, $\sigma^2_{\text{rev}} = 0.134$), and bank credit (`TOTBKCRNSA`, $\sigma^2_{\text{rev}} = 0.133$) exhibit substantial post-release noise. Initial survey estimates are routinely revised as complete administrative datasets arrive.

For high-revision series, the forward-fill baseline suffers from double degradation: it projects obsolete time steps forward, and those historical steps are anchored to noisy initial figures.

5 Nowcasting

Every observation in a nowcasting pipeline is indexed by two distinct temporal dimensions:

1. **Event Time (t):** The reference period during which the economic activity actually occurred (e.g., July 2026).

2. **Vintage / Real-Time (τ):** The calendar date on which a specific data snapshot becomes known to the model.

Let $y_{i,t,\tau}$ denote the value of variable i for reference date t as known at vintage time τ. Because publication latency δ_i is substantial—ranging from 0.15 months for weekly jobless claims ('ICNSA') up to 1.55 months ($\sim$ 45 days) for housing starts ('HOUSTNSA') and 5.03 months for national debt ('GFDEBTN')—the current state vector at time τ contains an expanding rag-tag boundary of missing values where $t > \tau - \delta_i$.

$$\delta_i = \mathbb{E}[\tau_{\text{release}} - t]$$

5.1 Walk-Forward Validation Protocol

To evaluate out-of-sample nowcasting performance without lookahead bias or data contamination, predictions are generated through a rolling bitemporal walk-forward scheme:

1. **Window Setup:** At each walk iteration, a 6-year historical block is selected. The final 1 year of this block is reserved for validation, while the preceding 5 years are designated as the training period.

2. **Model Fitting:** A model is trained exclusively on the 5-year training period using only the data snapshot published as of the training cutoff date (1 year prior to the end of the window). All future data releases and retroactive revisions after this date are strictly masked.

3. **Monthly Stride Nowcasting:** Stepping month-by-month through the 1-year validation window:

   - **Point-in-Time Data Snapshot:** The dataset is updated to incorporate all published information available up to the current stride date.
   - **Inference:** The frozen model executes inference on this updated snapshot to generate a nowcast for the current stride date.

4. **Window Advancement:** Once monthly nowcasts are recorded across the entire 1-year validation period, the loop stops for that fold. The 6-year window then rolls forward by 1 year, ensuring every recorded nowcast is strictly out-of-sample without temporal overlap.

5.2 Bitemporal Benchmarking Framework

Evaluating whether a dynamic factor model provides genuine predictive skill requires isolating two distinct reference targets at every evaluation stride date τ. Because reporting agencies update historical numbers retroactively, evaluating a nowcast against a single static target introduces systemic bias. We define two evaluation targets to test distinct properties of the state-space model:

1. **Forward-Fill Baseline Target ($y_{i,t,\tau}$):** The most recent published figure available strictly at vintage time τ. This baseline represents zero-skill persistence: if no model were run, an analyst would assume the unobserved target t equals the latest available historical print $y_{i,t_0,\tau}$ (where $t_0 < t$).

2. **Oracle Ground-Truth Target ($y_{i,t,\infty}$):** The fully revised, terminal value known outside the walk-forward simulation window (as of maximum available revision history today). This represents the true, noise-free underlying economic state after all annual benchmark revisions and seasonal re-estimations have settled.

Comparing a model nowcast $\hat{y}_{i,t|\tau}$ against both targets allows us to measure two critical capabilities. First, relative to the *Forward-Fill Baseline*, it measures whether the factor space extracts useful leading information from high-frequency co-moving series during publication gaps. Second, relative to the *Oracle Target*, it measures the model's ability to filter out transient measurement noise present in initial agency announcements.

6 Time Series Stationarity

To ensure valid state-space estimation and prevent spurious correlation in the dynamic factor model, raw macroeconomic time series $x_{i,t}$ are transformed into stationary representations

$y_{i,t} = T_{c_i}(x_{i,t})$ following the standardized FRED-MD transformation codes $c_i \in \{1, 2, \dots, 7\}$ McCracken and Ng [2015], with the addition of a custom $c_i \in \{8, 9, 10\}$ to handle semantic units. While Transformations 10 could be collapsed directly to transformation 1, and transformations 8,9 can be collapsed into transformation 2, augmentation was created to accomodate for advanced feature augmentation.

6.1 Forward Stationarity Transformations

Let x_t denote the raw observable level at time t, and let $\Delta x_t = x_t - x_{t-1}$ denote the first difference operator. The forward transformation operator $T_c(x_t)$ maps non-stationary raw inputs into stationary inputs y_t:

$$y_t = T_c(x_t) = \begin{cases} x_t & \text{if } c = 1 \quad \text{(Level)} \\ \Delta x_t & \text{if } c = 2 \quad \text{(First Difference)} \\ \Delta^2 x_t = x_t - 2x_{t-1} + x_{t-2} & \text{if } c = 3 \quad \text{(Second Difference)} \\ \ln(x_t) & \text{if } c = 4 \quad \text{(Natural Logarithm)} \\ \Delta \ln(x_t) = \ln(x_t) - \ln(x_{t-1}) & \text{if } c = 5 \quad \text{(First Difference of Natural Log)} \\ \Delta^2 \ln(x_t) = \ln(x_t) - 2\ln(x_{t-1}) + \ln(x_{t-2}) & \text{if } c = 6 \quad \text{(Second Difference of Natural Log)} \\ \Delta \left(\frac{x_t - x_{t-1}}{x_{t-1}} \right) & \text{if } c = 7 \quad \text{(First Difference of Percent Change)} \\ \frac{\Delta x_t}{100} & \text{if } c = 8 \quad \text{(Scaled Difference)} \\ \frac{\Delta x_t}{D} & \text{if } c = 9 \quad \text{(Custom Denominator Difference)} \\ \frac{x_t}{100} & \text{if } c = 10 \quad \text{(Fractional Level)} \end{cases} \tag{5}$$

where $D > 0$ represents a scale adjustment factor (e.g., $D = 10^{12}$ for trillion-dollar national debt series).

6.2 Reverse Level Reconstruction

For downstream evaluation, reporting, and level nowcasting, stationary factor projections $\hat{y}_t$ are mapped back into raw level space $x_t = T_c^{-1}(y_t, x_{t_0})$ using seed values x_{t_0} anchoring the initial state at reference point $t_0 < t$:

$$x_t = T_c^{-1}(y_t) = \begin{cases} y_t & \text{if } c = 1 \\ x_{t_0} + \sum_{k=t_0+1}^{t} y_k & \text{if } c = 2 \\ x_{t_0} + (t - t_0)(x_{t_0} - x_{t_0-1}) + \sum_{k=t_0+1}^{t} \sum_{j=t_0+1}^{k} y_j & \text{if } c = 3 \\ \exp(y_t) & \text{if } c = 4 \\ x_{t_0} \cdot \exp\left(\sum_{k=t_0+1}^{t} y_k\right) & \text{if } c = 5 \\ \exp\left(\ln(x_{t_0}) + (t - t_0)\Delta \ln(x_{t_0}) + \sum_{k=t_0+1}^{t} \sum_{j=t_0+1}^{k} y_j\right) & \text{if } c = 6 \\ x_{t_0} \prod_{k=t_0+1}^{t} \left(1 + r_{t_0} + \sum_{j=t_0+1}^{k} y_j\right) & \text{if } c = 7 \\ x_{t_0} + 100 \sum_{k=t_0+1}^{t} y_k & \text{if } c = 8 \\ x_{t_0} + D \sum_{k=t_0+1}^{t} y_k & \text{if } c = 9 \\ 100 \cdot y_t & \text{if } c = 10 \end{cases} \tag{6}$$

where $r_{t_0} = \frac{x_{t_0} - x_{t_0-1}}{x_{t_0-1}}$ denotes the baseline rate of change prior to the unobserved forecasting horizon.

6.3 Augmented Dickey-Fuller (ADF) Stationarity Analysis

To determine whether a raw time series x_t requires differencing prior to state-space factor estimation, we evaluate unit root dynamics using the Augmented Dickey-Fuller (ADF) test framework Dickey and Fuller [1979, 1981]. For a given time series sample of length N, the test regression with an autoregressive order p is specified as:

$$\Delta x_t = \alpha + \beta t + \gamma x_{t-1} + \sum_{j=1}^{p} \delta_j \Delta x_{t-j} + \varepsilon_t$$

where Δ is the first-difference operator, α is a constant drift term, t is a deterministic time trend, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ is white noise error. The lag length p is dynamically selected to minimize the Akaike Information Criterion (AIC):

$$\text{AIC}(p) = 2k - 2\ln(\hat{L})$$

The null hypothesis $H_0 : \gamma = 0$ implies the presence of a unit root (non-stationary), whereas the alternative hypothesis $H_1 : \gamma < 0$ implies stationary mean reversion.

A series is classified as statistically stationary if its asymptotic p-value satisfies $p \leq \alpha$, where the critical significance threshold is set to $\alpha = 0.05$. To prevent sample distortion, two deterministic evaluation guardrails are enforced prior to running the test regression:

1. **Sample Size Constraint:** If the valid non-null observation count satisfies $N < 12$, the series is defaulted to stationary level ($c = 1$) due to insufficient statistical power.
2. **Degenerate Variance Guard:** If the sample variance satisfies $\text{Var}(x_t) \leq \epsilon$, the series contains no dynamic variation and is automatically classified as stationary ($c = 1$).

6.4 Semantic-Augmented Stationarity Classification

Purely statistical Augmented Dickey-Fuller (ADF) tests frequently fail on macroeconomic time series due to structural breaks, finite-sample bias, and bounded support near zero bounds (e.g., policy rates or unemployment figures) Dickey and Fuller [1979, 1981]. To overcome these statistical edge cases, our framework combines empirical ADF unit-root testing with a domain-aware semantic classifier that parses series metadata, including title strings, measurement units, and reporting notes.

The hybrid classification pipeline evaluates candidate time series through a three-stage decision hierarchy:

1. **Rule-Based Domain Overrides:** Explicit series mappings and regular expression patterns take precedence over purely statistical tests:
   - **Flow and Balance Sheet Scaling ($c = 9$):** High-magnitude monetary flows (surpluses, deficits, net issuances) are assigned custom scale denominators $D \in \{10^3, 10^6, 10^9\}$ based on reported monetary units to maintain numerical stability without destroying additive linear properties.
   - **Volatility and Index Bounds ($c = 5$):** Bounded positive financial market indices and volatility measures (e.g., VIX) are mapped to log first-differences ($c = 5$) when strictly positive, or linear first-differences ($c = 2$) if zero-crossings occur.
   - **Survey Balances and Spreads ($c = 8$):** Net percentage balances, spreads, and survey diffusion metrics are transformed using scaled linear differences ($\frac{\Delta x_t}{100}$).
2. **Mathematical Transformation Guardrails:** Structural safety constraints enforce domain boundaries prior to statistical testing:

- **Rate and Percentage Guard:** Series expressed as percentage rates, interest rates, or ratios are forbidden from logarithmic transformations to prevent negative domain errors ($\ln(x \le 0)$) and non-linear distortion.
- **Log Positivity Constraint:** A natural log transformation is permitted only if $\min(x_t) > 10^{-6}$.
- **Price Index Constraint:** Second-difference log transformations ($c = 6$) are restricted strictly to explicit price deflators and price indices (e.g., CPI, PPI, PCE).

3. **Sequential Statistical Fallback Structure:** Unmatched series fall back to a constrained, ordered ADF testing sequence:
   - **Level Test ($c = 1$):** Raw levels are tested for stationarity. If $p \le 0.05$ and the series is not an unscaled nominal aggregate (e.g., raw population or dollar levels), the level form ($c = 1$) is retained.
   - **Log Difference Test ($c = 5, 6$):** For positive non-rate aggregates, log first-difference $\Delta \ln(x_t)$ is evaluated. If unit-root non-stationarity persists and the series is a recognized price index, log second-difference $\Delta^2 \ln(x_t)$ is evaluated.
   - **Linear Difference Test ($c = 2, 3$):** For bounded rates or non-positive series, first difference Δx_t is tested, falling back to second difference $\Delta^2 x_t$ if necessary.

6.4.1 Validation Against Benchmark Dataset (FRED-MD)

To evaluate the generalization and fidelity of our regex-driven semantic classifier, we benchmarked its predicted transformation codes (t-codes) against the canonical ground-truth annotations provided in the FRED-MD database McCracken and Ng [2015][1].

While standard FRED-MD applies seven traditional transformation codes ($c \in \{1, \dots, 7\}$), our system introduces three custom extension classes ($c \in \{8, 9, 10\}$) to handle high-magnitude flow scaling and bounded diffusion metrics cleanly. Evaluating across the complete FRED-MD panel of $N = 125$ macro series, our semantic classifier successfully reconstructed the benchmark transformation decisions with high precision:

Metric	**Count / Value**	**Percentage**
Total FRED-MD Series Evaluated (N)	125	100.0%
Exact Benchmark Matches	117	**93.6%**

Table 2: Classification accuracy and alignment of the semantic regex engine relative to the McCracken-Ng FRED-MD ground truth.

Achieving 93.6% alignment (117/125 series) demonstrates that our sequential regex hierarchy successfully reverse-engineers the expert heuristic rules governing macro stationarity transformations, providing an automated, deterministic pipeline for scaling large macro panels.

6.5 Publication Latency and Revision Variance Estimation

To configure state-space observation noise matrices in real time, empirical publication latency and historical revision variances are computed directly from point-in-time observation records.

Let $\mathcal{V}_i = \{(t, \tau, y_{i,t,\tau})\}$ denote the historical set of observed vintages for series i, where t is the reference event period and τ is the real-time publication date.

[1]Ground truth target retrieved from the FRED-MD vintage database (`2026-06-MD.csv`).

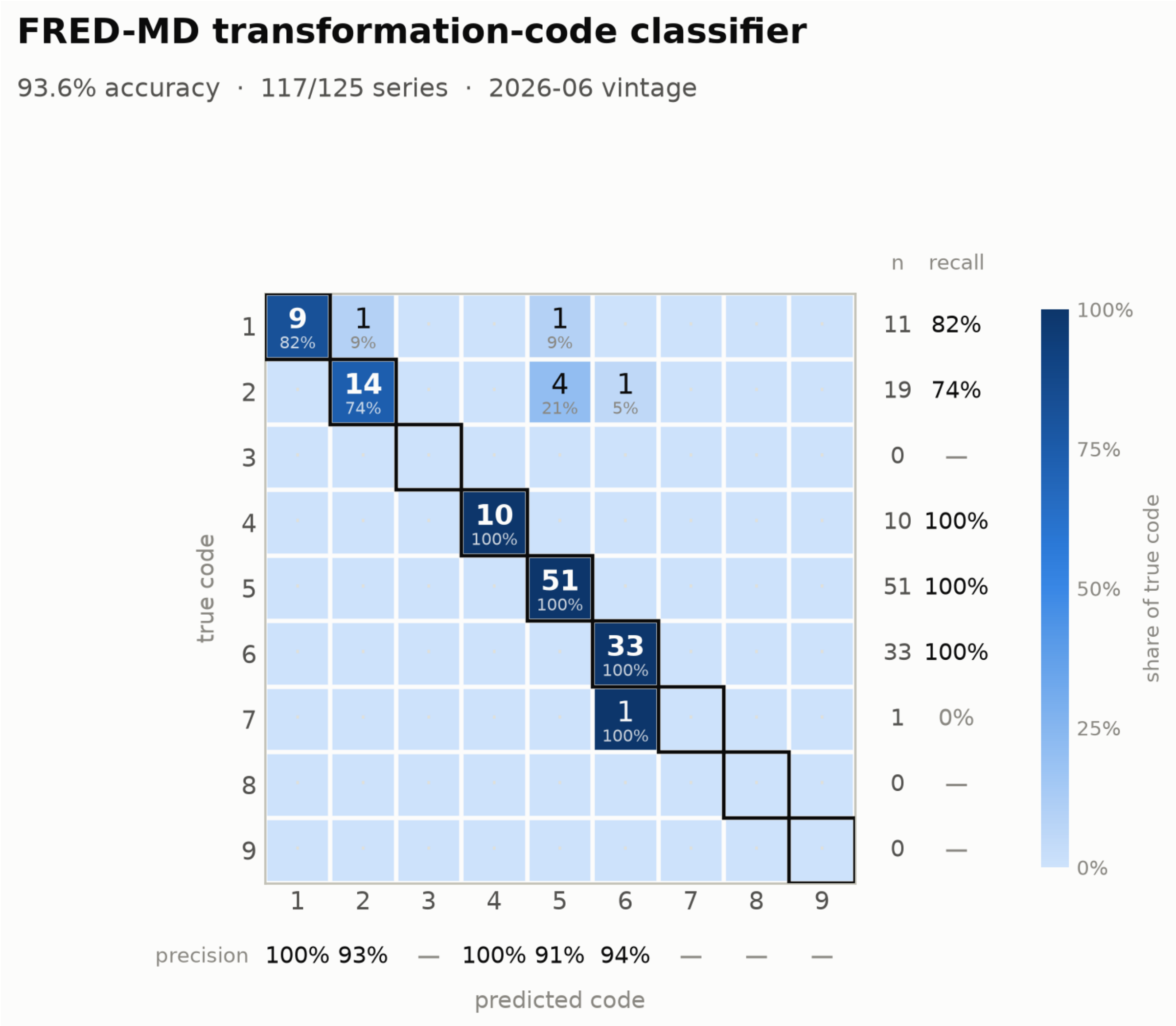

Figure 8: Confusion matrix illustrating classification performance of the custom semantic engine relative to the FRED-MD benchmark dataset (93.6% overall accuracy across 125 macroeconomic series).

6.5.1 Publication Latency Estimation

For each reference period t, the minimum publication latency in calendar days is defined relative to the earliest release vintage:

$$\tau_{\text{first}}(t) = \min\{\tau \mid (t, \tau) \in \mathcal{V}_i\}$$

$$\delta_{i,t} = \max\left(0, \frac{\tau_{\text{first}}(t) - t}{\Delta_{\text{resample}}}\right)$$

where Δ_{resample} represents the length of the resample frequency in days (e.g., $\Delta_{\text{resample}} = 30.4375$ for monthly data). The expected publication lag $\bar{\delta}_i$ and latency variance $\sigma^2_{\delta,i}$ are estimated as sample moments:

$$\bar{\delta}_i = \frac{1}{N_t}\sum_{t=1}^{N_t} \delta_{i,t}, \quad \sigma^2_{\delta,i} = \frac{1}{N_t - 1}\sum_{t=1}^{N_t}(\delta_{i,t} - \bar{\delta}_i)^2$$

To prevent database backfill artifacts (such as historical bulk data loads) from contaminating reporting latency statistics, mean latency is bounded by a frequency-dependent threshold C_{freq}:

$$C_{\text{freq}} = \begin{cases} 0.25 \text{ months} & \text{Daily (D)} \\ 0.50 \text{ months} & \text{Weekly (W)} \\ 3.00 \text{ months} & \text{Monthly (M)} \\ 6.00 \text{ months} & \text{Quarterly (Q)} \end{cases}$$

If the sample mean satisfies $\bar{\delta}_i > C_{\text{freq}}$, the latency is clipped to $\bar{\delta}_i = C_{\text{freq}}$ and the standard deviation is set to $\sigma_{\delta,i} = 0$.

6.5.2 Revision Error Variance Normalization

Revision error variance quantifies the noise present in initial agency announcements relative to fully settled historical figures. For every reference period t, we extract the initial published vintage $y_{i,t}^{\text{first}}$ and the terminal revised vintage $y_{i,t}^{\text{final}}$:

$$y_{i,t}^{\text{first}} = y_{i,t,\tau_{\text{first}}(t)}, \quad y_{i,t}^{\text{final}} = y_{i,t,\tau_{\max}(t)}$$

Both vintage vectors are transformed into stationary space using the assigned transformation code c_i:

$$w_{i,t}^{\text{first}} = T_{c_i}(y_{i,t}^{\text{first}}), \quad w_{i,t}^{\text{final}} = T_{c_i}(y_{i,t}^{\text{final}})$$

The raw revision error is defined as $e_{i,t}^{\text{rev}} = w_{i,t}^{\text{final}} - w_{i,t}^{\text{first}}$. To ensure scale-free variance injection into the Kalman filter observation error covariance matrix, the raw revision variance $\text{Var}(e_{i,t}^{\text{rev}})$ is normalized by the variance of the stationary series $\sigma_{w,i}^2$:

$$\sigma_{\text{rev},i}^2 = \frac{\text{Var}(e_{i,t}^{\text{rev}})}{\max(\sigma_{w,i}^2, 10^{-6})}$$

This normalized variance parameter $\sigma_{\text{rev},i}^2$ directly scales the observation noise covariance matrix R, instructing the Kalman filter to downweight initial releases proportional to their empirical revision volatility.

7 Kalman Filter

To estimate the latent macroeconomic factors from an unbalanced panel of mixed-frequency indicators, the dynamic factor model Kalman Filter Kalman [1960] is cast into a state-space representation. The system consists of two equations:

$$\text{Transition Equation (Dynamics):} \quad \mathbf{f}_t = A\mathbf{f}_{t-1} + \mathbf{u}_t, \quad \mathbf{u}_t \sim \mathcal{N}(0, Q) \tag{7}$$

$$\text{Observation Equation (Measurement):} \quad \mathbf{y}_t = \Lambda\mathbf{f}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(0, R) \tag{8}$$

where $\mathbf{f}_t$ is the unobserved latent factor state vector, $\mathbf{y}_t$ is the vector of observed macroeconomic series, A governs the autoregressive momentum of the economy, and Λ is the factor loading matrix mapping the latent economy to observable data. The covariance matrices Q and R represent systemic shocks and measurement error (such as agency revision variance), respectively.

7.1 The Kalman Filter Predict-Update Cycle

Because macroeconomic data suffers from publication lags (the "ragged edge"), $\mathbf{y}_t$ frequently contains missing values. The Kalman filter elegantly handles these gaps through a recursive, two-step Predict-Update cycle.

1. The Predict Step (Time Update): Given the factor estimate from the previous period, the filter projects the state of the economy forward using the transition dynamics:

$$\hat{\mathbf{f}}_{t|t-1} = A\hat{\mathbf{f}}_{t-1|t-1} \tag{9}$$

$$P_{t|t-1} = AP_{t-1|t-1}A^\top + Q \tag{10}$$

where P is the covariance matrix representing the model's uncertainty about the latent state. During periods with no new data releases (missing observations), the filter simply relies on this prediction step to traverse the unobserved horizon.

2. The Update Step (Measurement Update): When a new data release becomes available at time t, the filter calculates the forecast error (the "innovation" or "news") and updates its belief about the economy:

$$\mathbf{v}_t = \mathbf{y}_t - \Lambda\hat{\mathbf{f}}_{t|t-1} \quad \text{(Innovation)} \tag{11}$$

$$K_t = P_{t|t-1}\Lambda^\top(\Lambda P_{t|t-1}\Lambda^\top + R)^{-1} \quad \text{(Kalman Gain)} \tag{12}$$

$$\hat{\mathbf{f}}_{t|t} = \hat{\mathbf{f}}_{t|t-1} + K_t\mathbf{v}_t \tag{13}$$

$$P_{t|t} = (I - K_t\Lambda)P_{t|t-1} \tag{14}$$

7.2 The Role of the Kalman Gain (K_t)

The Kalman Gain K_t is the optimal weighting matrix that balances the model's internal prediction confidence against the external observation confidence. The observation noise matrix R is explicitly scaled by the empirically derived revision variances (σ^2_{rev}).

If a newly released indicator has high historical revision noise (large R), the Kalman Gain K_t shrinks toward zero. The filter treats the release as a noisy signal and relies more heavily on the model's autoregressive momentum $\hat{\mathbf{f}}_{t|t-1}$. Conversely, if a series has zero revision variance (e.g., financial market yields), K_t increases, forcing the latent state to aggressively snap to the observed data.

8 Mixed Frequency

8.1 Mixed-Frequency Alignment and Bitemporal State-Space Estimation

Macroeconomic time series operate across heterogeneous reporting frequencies (e.g., monthly employment figures vs. quarterly national accounts). Standard preprocessing methods that apply naive forward-filling (ffill) introduce zero-variance steps and artificial stationarity artifacts, distorting lead-lag dynamics. To preserve true temporal precedence without introducing lookahead bias, we deploy a two-stage mixed-frequency processing architecture.

8.1.1 Native-Frequency Transformation and Sparse Grid Alignment

In the initial feature engineering stage, raw observation contexts are aligned onto a uniform monthly grid $\mathcal{T}_{\text{monthly}}$. Series are categorized and transformed based on their underlying economic properties prior to panel aggregation:

1. **Flow vs. Stock Disambiguation:** Flow series (e.g., total trade volume) are aggregated within each frequency bin using a strict minimum observation count (min_count = 1) to

prevent unobserved periods from defaulting to zero. Stock and level variables (e.g., central bank assets) are averaged over their native observation windows.

2. **Native Stationarity Transformations:** Stationarity-inducing transformations—such as log-differencing or absolute differencing—are executed exclusively on the non-null, native observation sequence $Y_{i,\text{native}}$. For a quarterly series observed at periods t and $t-3$, the transformation
$$y_{i,t} = \log(Y_{i,t}) - \log(Y_{i,t-3}) \tag{15}$$
captures true quarter-over-quarter growth rather than multi-month zero-growth artifacts.

3. **Sparse Grid Re-indexing:** Transformed native series are re-aligned to $\mathcal{T}_{\text{monthly}}$. Off-month timestamps naturally remain missing values (NaN), avoiding synthetic growth spikes.

8.1.2 Dynamic Factor Estimation via State-Space Representations

To extract a continuous, high-frequency latent signal from the sparse, NaN-populated panel, we employ a Dynamic Factor Model in state-space form estimated via the Expectation-Maximization (EM) algorithm (DynamicFactorMQ):

$$\mathbf{y}_t = \mathbf{\Lambda}\mathbf{f}_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \mathbf{R}_t) \tag{16}$$
$$\mathbf{f}_t = \mathbf{A}_1\mathbf{f}_{t-1} + \cdots + \mathbf{A}_p\mathbf{f}_{t-p} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \mathbf{Q}) \tag{17}$$

where $\mathbf{y}_t$ is the $N \times 1$ vector of standardized mixed-frequency observations, $\mathbf{f}_t$ is the $K \times 1$ vector of latent unobserved factors, $\mathbf{\Lambda}$ represents the factor loading matrix, and $\mathbf{R}_t$ is the observation noise covariance matrix.

During the Kalman filtering pass, missing entries (NaNs) corresponding to off-month lower-frequency series are handled natively: the measurement update step dynamically collapses the observation vector $\mathbf{y}_t$ and corresponding rows of $\mathbf{\Lambda}$, updating the latent state $\mathbf{f}_t$ strictly using available higher-frequency prints.

8.1.3 Real-Time Nowcasting with Bitemporal Revision Uncertainty (R_t Penalty)

To account for measurement error in freshly released economic data subject to future revisions, we extend the state-space model by constructing a 3D time-varying observation covariance matrix $\mathbf{R}_t \in \mathbb{R}^{N\times N\times T}$.

For each series i observed at time t under a point-in-time evaluation boundary T_{now}, we compute the revision maturity τ_{rev} based on its initial release timestamp t_{first}:

$$\tau_{\text{rev}} = \max\left(\frac{T_{\text{now}} - t_{\text{first}}}{30.4375}, 0\right) \tag{18}$$

The diagonal elements of the observation covariance matrix $\mathbf{R}_{t,i,i}$ are then dynamically adjusted by penalizing fresh, unrevised observations:

$$\mathbf{R}_{t,i,i} = \mathbf{R}_{\text{base},i} + \sigma^2_{\text{rev},i} \cdot \exp\left(-\gamma_i \cdot \tau_{\text{rev}}\right) \tag{19}$$

where $\mathbf{R}_{\text{base},i}$ is the baseline variance, $\sigma^2_{\text{rev},i}$ represents the initial revision variance parameter, and γ_i dictates the exponential decay rate of uncertainty as the observation matures. Re-smoothing the state-space model over frozen MLE parameters with $\mathbf{R}_t$ yields real-time nowcasts that automatically downweight volatile, recently released indicators until they reach statistical maturity.

9 Sample Results

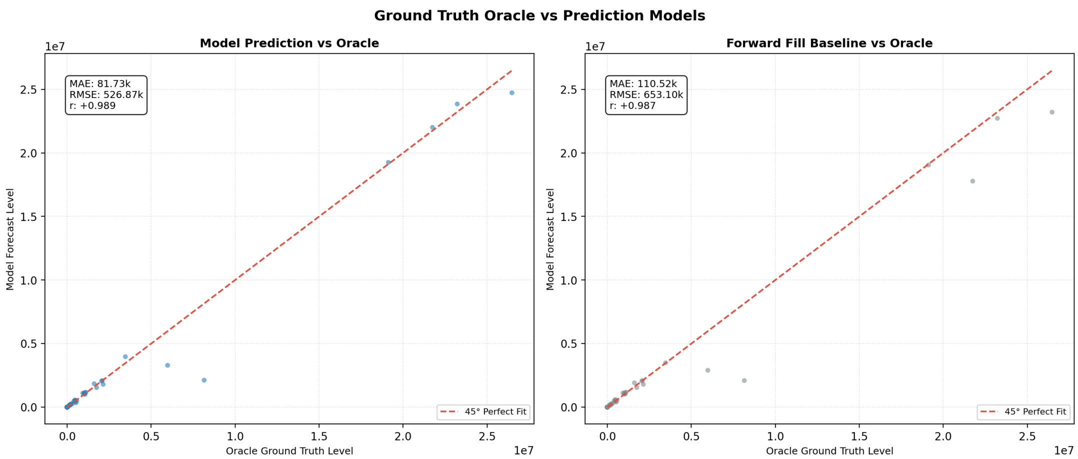

Figure 9: Error 527 RMSE Predicted vs 653 RMSE, Predicted Net +0.002 R^2

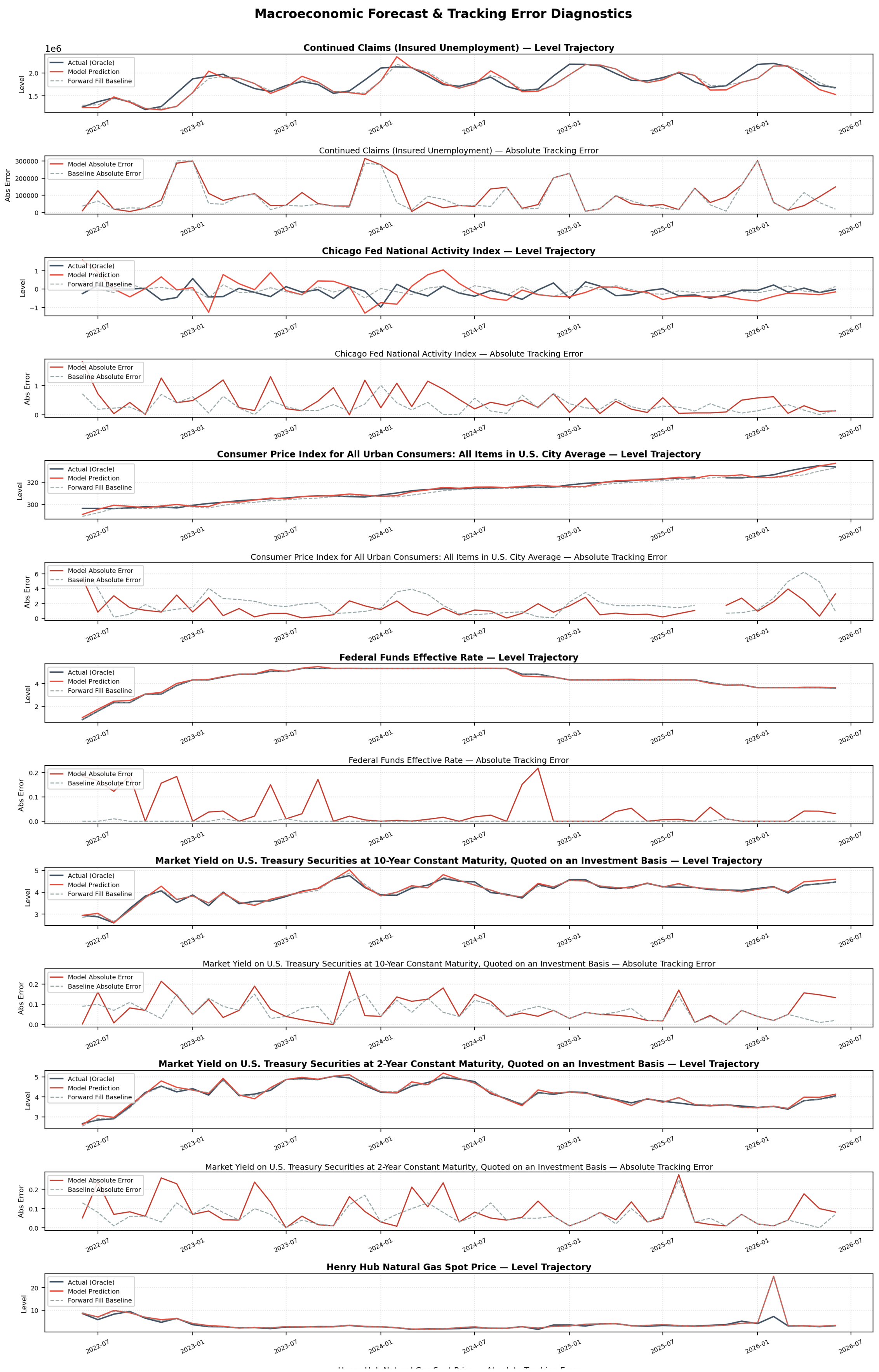

Figure 10: Global Timeline (Part 1 of 3)

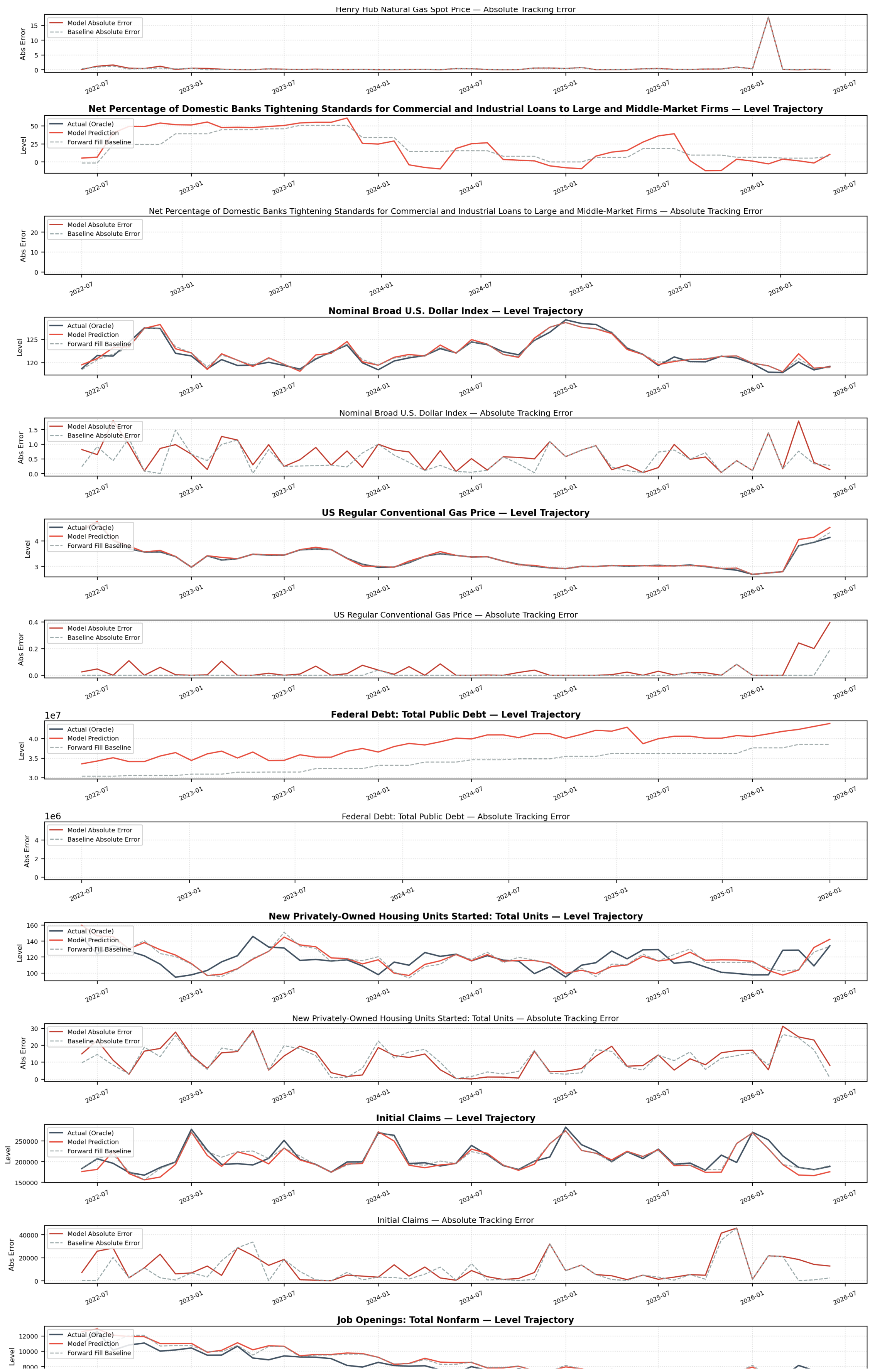

Figure 11: Global Timeline (Part 2 of 3)

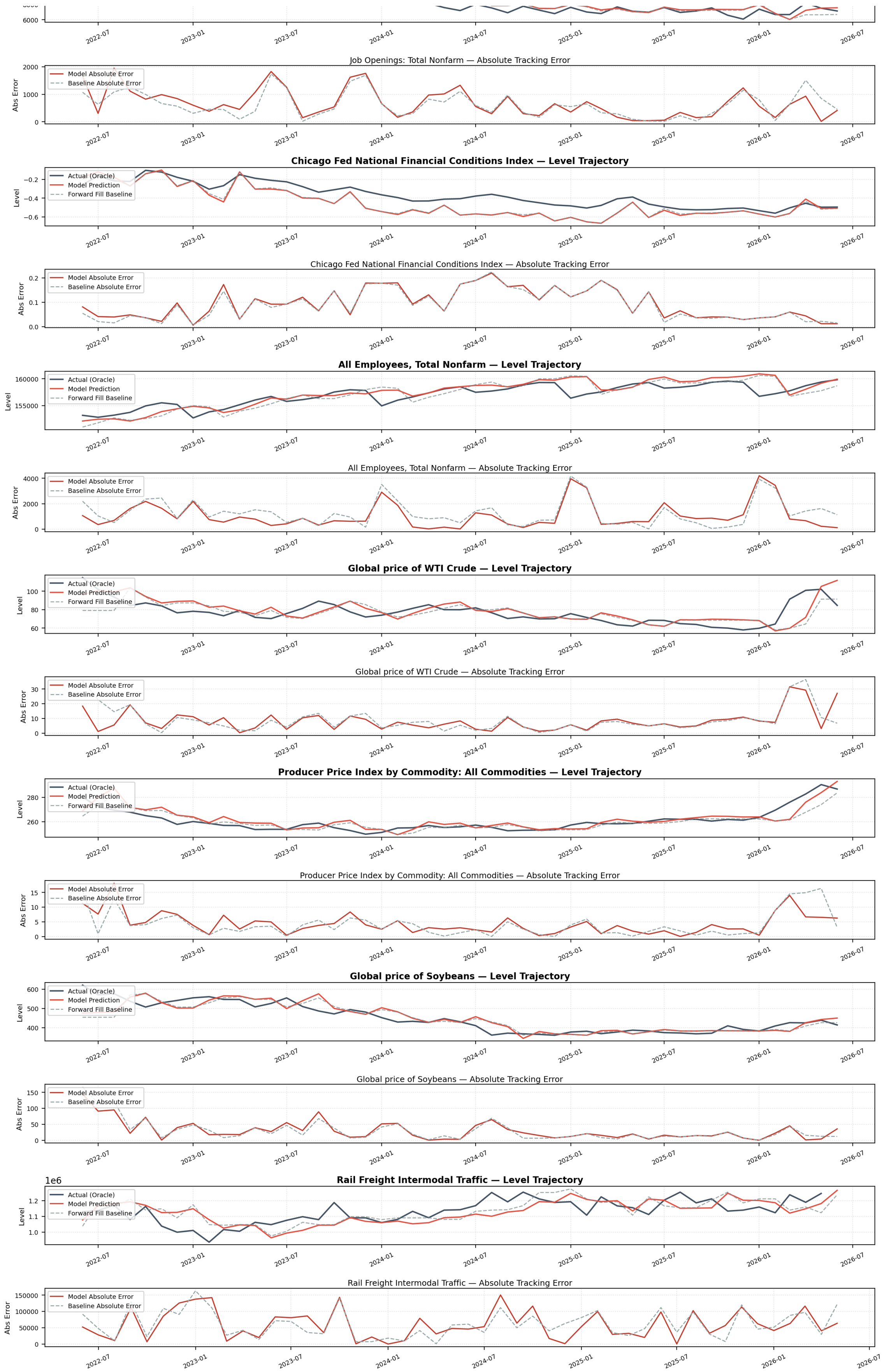

Figure 12: Global Timeline (Part 3 of 3)

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